HEMENDRA TRIPATHI

AI Voice Engineer | Technical Lead | Real-time voice agents & multi-LLM systems

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SUMMARY

AI voice engineer and Technical Lead with 4+ years shipping production AI systems. I lead engineering for Callin.io, a real-time voice AI platform serving 1,000+ businesses across healthcare and real estate. I own the full call path (SIP/VoIP and WebRTC orchestration, STT/LLM/TTS selection, multi-LLM routing, latency budget) and treat latency, cost and reliability as product features. Led a 5-engineer team from architecture to revenue.

TECHNICAL SKILLS

- **Voice & real-time AI:** SIP/VoIP, WebRTC, WebSockets, STT/TTS, voice cloning, ElevenLabs, Cartesia, Deepgram, Twilio, Telnyx, latency optimization
- **LLM systems:** multi-LLM routing, semantic caching, RAG (Pinecone, Supabase Vector), prompt engineering, self-hosted Qwen, post-call analytics
- **Platform:** Node.js, NestJS, Python, React, Next.js, React Native, PostgreSQL (Supabase), Redis, AWS, Docker, CI/CD, Stripe metering
- **Leadership:** architecture decisions, team leadership, hiring, mentoring (150+ students)

EXPERIENCE

Technical Lead, AI Voice Platform (Callin.io) | Appspundit Infotech, Udaipur, India | Feb 2026 – Present

- Own the real-time call path: SIP/VoIP and WebRTC orchestration, STT/LLM/TTS provider selection, and the end-to-end latency budget for voice agents.
- Built multi-LLM routing that sends each conversational turn to the smallest capable model, cutting inference cost by roughly 20% while keeping larger models for complex reasoning.
- Led the evaluation and switch to third-party voice providers (ElevenLabs and others), approved by the CEO; together with latency, STT/TTS, multi-language and agent-management improvements, this cut subscription cancellations by roughly 40%.
- Run self-hosted Qwen models for post-call analytics (goal analysis, sentiment, key moments, AI recommendations) and low-latency agent responses.
- Built an AI-agent ticket pipeline where changes are gated by blast radius, so low-risk fixes ship fast and risky ones get review.
- Extended the architecture to CondoMail (AI email triage and drafting) and Realead (AI lead-qualification calling), leading a 5-engineer team and reporting to the founder.

Software Engineer, AI Voice Platform (Callin.io) | Appspundit Infotech, Udaipur, India | Oct 2024 – Jan 2026

- Built Callin.io from the ground up as an AI telephony SaaS: voice agents, Twilio and Telnyx carrier integrations, multi-language agents, and Stripe usage-based billing.
- Designed the multi-tenant architecture on Supabase and PostgreSQL; worked directly with international clients on roadmaps and delivery.
- Implemented semantic caching and concurrent prompt execution to keep voice-agent response times low under load.

Full-Stack Developer (Freelance) | Remote | 2022 – 2024

- Built and shipped 10+ production web applications for clients across industries, including a FinTech management system that automated manual reconciliation. Handled architecture, backend, deployment and long-term maintenance.

Technical Instructor | Aimers Institute, Udaipur | Jul 2021 – Jul 2023

- Taught 150+ students Python, Django, MERN and Flutter through project-based learning; redesigned the curriculum around industry needs and supervised 30+ capstone projects.

PROJECTS

- **Callin.io:** Real-time AI voice platform (hemendra.readwith.io/work/callin-io). SIP/WebRTC, multi-LLM routing, self-hosted Qwen post-call analytics, usage-based billing. Serving 1,000+ businesses.
- **CondoMail:** AI agents that classify, draft and reply to email across providers; live in production with early adopters, including high-volume Amazon sellers.
- **Realead:** AI calling agent that qualifies and follows up with new property leads; in final beta.

EDUCATION

Bachelor of Computer Applications (BCA) | Mohanlal Sukhadia University, Udaipur | 2022

LANGUAGES

English (fluent), Hindi (native)